Edwin

Typed Solution Guide - Guide Mock 2

Mathematical symbols and equations have been set using LaTeX.

Item 1

Equation of the line AB

$$\vec{d} = \overrightarrow{AB} = \begin{pmatrix}5\\7\\4\end{pmatrix} - \begin{pmatrix}2\\3\\1\end{pmatrix} = \begin{pmatrix}3\\4\\3\end{pmatrix}$$ T-01

Therefore, the equation of the line is

equation of line

$$\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}2\\3\\1\end{pmatrix} + \lambda\begin{pmatrix}3\\4\\3\end{pmatrix}.$$ M-01

Distance from $C(1,-2,8)$ to AB

A general point on AB is

$$D(2+3\lambda,\ 3+4\lambda,\ 1+3\lambda).$$

Hence

Perpendicular vector

$$\overrightarrow{CD} = \begin{pmatrix}1+3\lambda\\5+4\lambda\\-7+3\lambda\end{pmatrix}.$$ M-01

02

Since the shortest distance is perpendicular to the line,

$$\overrightarrow{CD} \cdot \begin{pmatrix}3\\4\\3\end{pmatrix} = 0.$$

Thus

Usage of dot product M-01

$$3(1+3\lambda) + 4(5+4\lambda) + 3(-7+3\lambda) = 0,$$
$$3 + 20 - 21 + 9\lambda + 12\lambda + 9\lambda = 0, \qquad 2 + 30\lambda = 0,$$
$$\lambda = -\frac{1}{15}.$$ M-01

Value of λ

Therefore

$$\overrightarrow{CD} = \begin{pmatrix}1-\frac{3}{15}\\5-\frac{4}{15}\\-7+\frac{69}{15}\end{pmatrix} = \frac{1}{15}\begin{pmatrix}4\\71\\-36\end{pmatrix}.$$ M-01

The required distance is

8.6535

$$|CD| = \frac{1}{15}\sqrt{4^2 + 71^2 + (-36)^2} = 5.313 \text{ m}.$$ M-01

Distance 8.6535

Since 5.313 m < 9 m, the flood-light position is suitable and will not affect visibility of the patrol.

Decision → D1-01

Reading shelter

The quadratic road curve and the tangent line are used as follows:

$$y = 4 - x^2, \qquad y = mx + c.$$

At the point of contact,

$$4 - x^2 = mx + c,$$ T-01

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For tangency, the discriminant is zero: $x^2 + mx + (c-4) = 0$. M-01

$$m^2 - 4(c-4) = 0. \quad \text{M-01}$$

Using the given point $B(0,6)$ gives $c = 6$. Therefore

$$m^2 - 4(6-4) = 0, \quad m^2 = 8, \quad m = \sqrt{8}. \quad (\text{M-01})$$

The equation of the tangent is therefore

$$y = \sqrt{8}x + 6. \quad y = \pm x + 6 \quad \text{M-01}$$

Putting $y = 0$ gives

$$0 = \sqrt{8}x + 6, \quad x = -\frac{6}{\sqrt{8}}. \quad (\text{M-01})$$

Hence the shelter/tangent position is obtained from the intercept values. D-01

08

Camera coverage

The coverage is modelled by

$$D = 20\,[3\sin\theta + 4\cos\theta]\,.$$

Using the R-formula,

$$3\sin\theta + 4\cos\theta = R\sin(\theta + \alpha), \quad \text{T-01}$$

where

$$R = \sqrt{3^2 + 4^2} = 5, \qquad \tan\alpha = \frac{4}{3}. \quad \text{M-01} \quad \text{either R or tan ---}$$

Thus

$$D = 100\sin(\theta + \alpha). \quad \text{M-01}$$

$+120$

The maximum possible value is

$$D_{\max} = 100 \text{ m}. + 120 = 220\text{m} \quad \text{M-01}$$

This is greater than the required 80 m, so the camera range is ~~adequate~~. inadequate.

D-01

pH

20

Item 2

Let x be the cost of layers' mash, y the cost of maize bran and z the cost of protein concentrate per 10 kg.

$$3x + y + 2z = 280000, \quad (1)$$
$$x + 2y + z = 180000, \quad (2)$$
$$2x + y + 3z = 260000. \quad (3)$$

T-01 (any one correct equation)

Eliminating z:

$$(1) - 2(2): \quad x - 3y = -80000,$$
$$3(1) - (3): \quad x + 5y = 280000.$$

M-01 (any one correct equation with x)

Solving,

$$8y = 360000, \quad y = 45000,$$
$$x = 55000, \quad z = 35000.$$

M-01 (any one correct value of x, y or z)

Testing the friendliness of the costs:

$$A = 4x + 4y + 3z.$$

Substituting,

$$A = 4(55000) + 4(45000) + 3(35000) = 505000.$$

Since

$$505000 < 550000,$$

M01 or ~~505,000 < 550,000~~ ~~5050~~

the feeds are cost-friendly.

D-01

Vitamin boost vs egg production

$$\sqrt{3x + 700} + \sqrt{2x + 600} = 70.$$

T-01

Squaring and simplifying gives

$$3x + 700 = 4900 + 2x + 600 - 140\sqrt{2x + 600},$$
$$x - 4800 = -140\sqrt{2x + 600}.$$

M-01

Squaring again,

$$x^2 - 9600x + 23040000 = 19600(2x + 600),$$
$$x^2 - 9700x + 28200000 = 0,$$
$$(x - 9400)(x - 300) = 0.$$

M-01 (any method of solving)

Testing in the original equation shows that

$x = 9400$ or $x = 300$. M-01

$\therefore x = 300$ M-01

Hence

$$\sqrt{1600} + \sqrt{1200} = 40 + 30 = 70.$$

The vitamin boost is successful. D-01

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Storage facility

$130n \geq n^2 + 3000$ (T-01)

$n^2 - 130n + 300 < 0$

The storage capacity is modelled by $N = (2+x)^{10}$.

$(n-30)(n-100) < 0$

Using binomial expansion,

$$N = 2^{10} + \binom{10}{1}2^9 x + \binom{10}{2}2^8 x^2 + \binom{10}{3}2^7 x^3 + \cdots.$$

For critical values $n = 30$ or 100

For $x = 0.5$,

$$N \approx 1024 + 5120(0.5) + 11520(0.5)^2 + 15360(0.5)^3,$$

$N = 3264$ hundreds of boxes $= 326400$ boxes.

Therefore there is no need for expanding the storage facility because the available storage is enough.

Criterion A - Item 2

Code	Criterion	Score
T21	Interprets simultaneous equations	1
T22	Interprets the cost friendliness of feeds	1
T23	Interprets the storage statement	1
T24	Interprets using binomial expansion	1
M21	Forms three simultaneous equations	1
M22	Eliminates one variable consistently	1
M23	Obtains cost of one component	1
M24	Obtains all component costs	1
M25	Substitutes into total cost	1
M26	Calculates final cost correctly	1
M27	Sets up vitamin equation	1
M28	Squares correctly	1
M29	Forms quadratic equation	1
M210	Solves quadratic equation	1
M211	Tests $x = 300$ in original equation	1
M212	Factors required polynomial	1
M213	Gives critical values	1
M214	Constructs sign table	1
M215	Interprets storage inequality	1
M216	Correct expansion and conclusion	1
D21	Correct decision on cost of feeds	1
D22	Correct decision on vitamin mix	1
D23	Correct decision on storage facility	1

Total: 23 marks

$n<30$	$30<n<100$	$n>100$
+	−	+

(M-01)

(M-01)

Since ~~Here~~ $30<n<100$, contains 33, the batch can be cleared for immediate sale. (D-01)

15

The volume is given by

$$V = ax^2 + bx^2 + cx + 30.$$

Using the given values:

$$V(1) = 28, \quad V(2) = 22, \quad V(3) = 18.$$

This gives

$$\left.\begin{aligned} a + b + c &= -2, \\ 4a + 2b + c &= -4, \\ 9a + 3b + c &= -4. \end{aligned}\right\} \quad \text{T-01 (any one equation)} \qquad \begin{aligned} &(1)\\ &(2)\\ &(3) \end{aligned}$$

Subtracting,

$$\begin{aligned} (2) - (1): &\quad 3a + b = -2, \\ (3) - (2): &\quad 5a + b = 0. \end{aligned} \quad \text{M-01 (an equation with two variables)}$$

Therefore

$$b = -5a, \quad 3a - 5a = -2, \quad a = 1.$$

Thus

$$b = -5, \quad c = 2. \quad \text{(M-01) one of the values of a, b or c}$$

Hence

$$V = x^3 - 5x^2 + 2x + 30. \quad \text{M-01 - The expression for V;}$$

When the temperature is 45°C, $x = 4.5$:

$$V = (4.5)^3 - 5(4.5)^2 + 2(4.5) + 30 = 28.875. \quad \text{M-01}$$

28.875 > all other values

The volume is greater than the volume in the table; therefore the specialist's advice is correct. D-01

Machine workload

$$2^{2t+2} = 33(2^t) - 8. \quad \text{T-01}$$

Let

$$2^t = k.$$

Then

$$\begin{aligned} 2^2(2^t)^2 - 33(2^t) + 8 &= 0, \\ 4k^2 - 33k + 8 &= 0, \quad \text{M-01} \\ (4k - 1)(k - 8) &= 0. \end{aligned} \quad \text{(M-01 or Substituting in bulldozer)}$$

Thus

$$k = \frac{1}{4} \quad \text{or} \quad k = 8. \quad \text{~~M-01 (Normal solving) or Sub~~}$$

Therefore

$$\begin{aligned} 2^t = 2^{-2} \quad &\text{or} \quad 2^t = 2^3, \\ t = -2 \quad &\text{or} \quad t = 3. \end{aligned} \quad \text{M-01}$$

Since operational time must be positive and less than 5 hours,

$$t = 3 < 5.$$

The operational time is good enough and the specialist's advice is valid. D-01

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Storage facility

$$N = 2^{10} + \binom{10}{1} 2^9 x^0 + \binom{10}{2} (2^8) x^2 + \cdots \quad \text{T-01}$$

$$N = 2^{10} + \binom{10}{1} 2^9 (0.25) + \cdots \quad \text{M-01}$$

$$= 3264 \times 100$$

$$= 326,400 \text{ boxes} \quad \text{M-01}$$

$326400 > 300000$, the storage space is insufficient A-01

Item 4

Container/tin enlargement

For a cylindrical tin,

$$V = \pi r^2 h, \qquad h = 3r.$$

Therefore

$$V = 3\pi r^3. \quad \text{—— T-01}$$

Differentiating,

$$\frac{dV}{dr} = 9\pi r^2. \quad \text{M-01}$$

For small changes,

$$\Delta V \approx 9\pi r^2 \Delta r.$$

Hence

$$\frac{\Delta V}{V} = \frac{9\pi r^2 \Delta r}{3\pi r^3} = 3\frac{\Delta r}{r}. \quad \text{m-01}$$

Given $\Delta r/r = 0.5\%$,

$$\frac{\Delta V}{V} = 3(0.5\%) = 1.5\% < 2\%. \quad \text{m-01}$$

Therefore the farmer should accept the tins. D-01

Pump rate

The height is

$$h = \frac{t^2}{5} + 3t.$$

Hence

$$\frac{dh}{dt} = \frac{2t}{5} + 3. \quad \text{T—01}$$

At $t = 3$,

$$\frac{dh}{dt} = \frac{6}{5} + 3 = 4.2. \text{ cm/s} \quad \text{(m-01)}$$

The volume is

$$V = \frac{t^2 + 4t}{t + 1}.$$

Differentiating,

$$\frac{dV}{dt} = \frac{(2t+4)(t+1) - (t^2+4t)(1)}{(t+1)^2}.$$

At $t = 3$,

$$\frac{dV}{dt} = \frac{(10)(4) - (21)(1)}{4^2} = \frac{19}{16}. \text{ cm}^3/\text{s} \quad \text{m-01}$$

Using the chain rule,

$$\frac{dV}{dh} = \frac{dV}{dt} \cdot \frac{dt}{dh} = \frac{19}{16} \times \frac{5}{21} = 0.2827 \text{ cm}^2. \quad \text{m-01}$$

Since this is less than 0.3, there will be no spillage and the farmer should use it. D-01

The farmer should use the machine

The rate of change of power is

Therefore $\frac{dp}{dt} = 20e^{-t/5}$.

$$\int dp = \int 20e^{-t/5}\,dt,$$

$$p = -100e^{-t/5} + C.$$

When $t = 0$, $p = 0$, so

Thus $0 = -100 + C, \quad C = 100.$

When $t = 5$ weeks, $p = -100e^{-t/5} + 100.$

The cost is $p = -100e^{-1} + 100 = 63.2121.$

$63.2121 \times 1000 = 63212.10$ shillings.

Since this is less than 140000, the farmer should consider this source of power.

$\frac{dp}{dt} = 20te^{-t/5}$ (T-01) ✓

$\int dp = 20\int te^{-t/5}$ for by parts (M-01) ✓

$p = -100(t+5)e^{-t/5} + C$ (M-01) ✓

when $t=0$, $p=0$ (initial conditions)

$0 = -100(0+5)e^{-0/5} + C$

$C = 500$

$p = 500 - 100(t+5)e^{-t/5}$

when $t=5$,

$p = 500 - 100(5+5)e^{-5/5}$

$p = \text{Shs. } 132.12$ (-M-01)

The power cost is 132.12 × 1000

$= 132120$

$132120 < 140,000$, the farmers should consider this source of power (D-01)

05

Criterion A - Item 4

Code	Criterion	Score
T41	Interprets $r = 3$ or $V = 3\pi r^3$	1
T42	Interprets small changes	1
T43	Interprets exact rate	1
M41	Finds dV/dr	1
M42	Relates ΔV with small changes	1
M43	Finds percentage trend of V and r	1
M44	Substitutes for $\Delta V/V$	1
M45	Gives conclusion for tins	1
M46	Finds dh/dt	1
M47	Finds dh/dt at $t = 3$	1
M48	Finds dV/dt	1
M49	Finds dV/dt at $t = 3$	1
M410	Uses chain rule	1
M411	Finds dV/dh at $t = 3$	1
M412	Integrates dp/dt	1
M413	Uses initial condition to find C	1
M414	Finds p at $t = 5$	1
M415	Calculates shillings	1
M416	Gives valid conclusion	1
D41	Correct decision on tins	1
D42	Correct decision on pump rate	1
D43	Correct decision on power cost	1

Total: 22 marks

For ladder method of differentiation or integration (M-01)

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Item 5

Error in height of the tank

The height is

$$h = \tan\theta.$$

Differentiating,

$$\frac{dh}{d\theta} = \sec^2\theta. \quad (T-01)$$

For a small change,

$$\Delta h \approx \sec^2\theta\,\Delta\theta.$$

At $\theta = 45°$,

$$\Delta\theta = 50^\circ - 45^\circ = 5^\circ = \frac{5\pi}{180 \times 60} = \frac{5\pi}{2160}. \quad (M-01)$$

Therefore

$$\Delta h = \sec^2 45^\circ \left(\frac{5\pi}{2160}\right) = 2\left(\frac{5\pi}{2160}\right) = 0.00291 \text{ m} < 0.2 \text{ m}.$$

Thus the school administration should buy or accept the tank. D-01

Water pump

$$\frac{dV}{dt} = \frac{100t}{(t+1)(t+2)}.$$

Thus

$$\int dV = \int \frac{100t}{(t+1)(t+2)}\,dt. \quad T-01 \text{ (Separating Variables)}$$

Using partial fractions,

$$\frac{100t}{(t+1)(t+2)} = \frac{A}{t+1} + \frac{B}{t+2}. \quad T-01 \text{ (Partial fractions)}$$

Then

$$100t = A(t+2) + B(t+1).$$

Putting $t = -1$ gives

$$-100 = A, \quad m-01 \text{ (any one value)}$$

and putting $t = -2$ gives

$$-200 = -B, \qquad B = 200.$$

Therefore

$$V = \int \frac{200}{t+2}\,dt - \int \frac{100}{t+1}\,dt,$$

$$V = 200\ln(t+2) - 100\ln(t+1) + C. \quad m-01$$

When $t = 0$, $V = 0$:

$$0 = 200\ln 2 + C, \qquad C = -200\ln 2.$$

Hence

$$V = 200\ln(t+2) - 100\ln(t+1) - 200\ln 2.$$

When $t = 5$,

$$V = 200\ln 7 - 100\ln 6 - 200\ln 2 = 73.37 \text{ m}^3. \quad m-01$$

Since $73.37 \text{ m}^3 > 70 \text{ m}^3$, the pump provides enough water and should be purchased. D-01

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$\Delta\theta = \frac{5}{180}\pi$ (M-01)

$\Delta h = \sec^2 45 \left(\frac{5\pi}{180}\right)$

$\Delta h = 0.1745 \text{ m}$

$0.1745 \text{ m} < 0.2 \text{ m}$

The school admin should buy the tank accept

04

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Chemical treatment

The rate of change of chemical mass is

$$\frac{d}{dt}(me^{-t}) = t^2.$$

Integrating,

$$me^{-t} = \frac{t^3}{3} + C.$$

When $t = 0$, $m = 100$:

$$100 = C.$$

Thus

$$me^{-t} = \frac{t^3}{3} + 100.$$

At $t = 6$,

$$me^{-2} = \frac{216}{3} + 100,$$

$$m = \left(\frac{8}{3} + 100\right)e^2 = 758.7\text{ kg} > 0\text{ kg}.$$

Therefore the chemical is safe and valid.

Criterion A - Item 5

Code	Criterion	Score
T51	Interprets small changes	1
T52	Interprets water volume/pump	1
T53	Interprets exact rate/chemical treatment	1
M51	Finds $dh/d\theta$	1
M52	Relates small changes	1
M53	Finds $\Delta\theta$	1
M54	Substitutes for Δh	1
M55	States conclusion on tank	1
M56	Uses partial fractions	1
M57	Finds constants A and B	1
M58	Integrates correctly	1
M59	Uses initial condition	1
M510	Finds volume at required time	1
M511	States correct pump conclusion	1
M512	Sets up chemical treatment equation	1
M513	Integrates correctly	1
M514	Uses initial condition	1
M515	Evaluates chemical mass	1
M516	Gives conclusion	1
D51	Correct decision on tank	1
D52	Correct decision on pump volume	1
D53	Correct decision on chemical treatment	1

Total: 22 marks

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